

David Silva

SENIOR SOFTWARE ENGINEER

+44 7427 910091 · davidslv.london@gmail.com · linkedin.com/in/davidslv · github.com/davidslv · davidslv.uk · London, UK

PROFESSIONAL SUMMARY

Senior software engineer with 15+ years building scalable, distributed systems serving millions of users across government, fintech, and high-growth startups. Deep expertise in Ruby and Rails, event-driven architecture, AI integration, and platform reliability at scale. Currently working across the architecture, observability, AI, and reliability of an FCA-regulated mortgage platform processing 1.7M requests a day. I build systems that **earn their complexity through evidence, not fashion**. Proven track record of measurable impact, including an **80% reduction in support tickets** and customer-facing AI on platforms serving **1M+ users**.

KEY SKILLS

LANGUAGES & FRAMEWORKS	Ruby/Rails, JavaScript, SQL, Python
ARCHITECTURE	Distributed Systems, Event-Driven Architecture, Modular Monoliths / Rails Engines, Microservices, REST APIs
DATA & MESSAGING	PostgreSQL, MySQL, Redis, Cassandra, Sidekiq, Kafka, Elasticsearch, Snowflake
CLOUD & AI	AWS (EC2, S3, Lambda, RDS), AWS Bedrock, LLM integration
OBSERVABILITY & DEVOPS	NewRelic, Datadog, Sentry, Docker, GitHub Actions, CircleCI
PRACTICES	TDD, Pair Programming, Agile/Scrum, Technical Mentoring, Onboarding

PROFESSIONAL EXPERIENCE

Senior Software Engineer, Tembo

Feb 2026 – Present

Working across the architecture, observability, AI, and reliability of an FCA-regulated mortgage platform handling ~1.7M requests and 500K+ background jobs a day across 8 services. Ruby, Rails, PostgreSQL, Redis, Sidekiq, AWS, Bedrock, NewRelic, Sentry, Docker.

- Built and productionised a **unified case-management tool** that replaced the 22 separate systems mortgage brokers consulted per case with one centralised workflow, cutting case-handling time from **~65 minutes to ~10 minutes** so brokers can advise applicants live on a call — designed to double mortgage capacity without added headcount.
- Engineered **AI bank-statement analysis on AWS Bedrock** to extract income and risk signals from PDF statements for mortgage fact-finding, solving a token-limit failure that silently dropped transactions and contributing the Bedrock provider upstream to the **ActiveAgent** open-source project.
- Leading a **third-party integration** that automates lender application submission so brokers stop re-keying data by hand, owning it end to end from partner evaluation and mapping 1,100+ data fields through to the build.
- Established **platform observability from scratch** with six NewRelic dashboards, including the board-level OKR instrument the company steers by, surfacing systemic issues such as an integration at ~40% success generating ~17,000 dead background jobs a week.
- Diagnosed and recovered a **partner sync silently broken for a month**, re-architecting it to per-user fan-out with retry resilience and landing the first clean overnight run at 14,213 users.
- Set architecture direction through a full platform review, cut the **deployment pipeline from ~2 hours to 35 minutes**, and upgraded Ruby 3.2 to 3.4 ahead of EOL.

Backend Senior Software Engineer, Indeed

2022 – 2025

Led backend engineering initiatives on a platform serving 1M+ workers across US/UK markets. Ruby, Rails, PostgreSQL, Redis, Kafka, OpenAI.

- Transformed the penalty appeals system over 6 months with cross-functional product teams, **reducing support tickets by 80%** and eliminating wrongful bans through system redesign and improved logic.
- Led rapid response to **critical incidents affecting 100K+ workers**, coordinating data-migration crisis recovery and preventing platform disruption.
- Collaborated on an **AI chat interface** using OpenAI GPT and custom few-shot learning, enabling natural-language task completion on job-posting tasks.
- **Mentored engineers** through pair programming, 1-on-1s, and team-wide knowledge sharing, and onboarded new hires across all levels.

Senior Software Engineer, Ministry of Justice

2020 – 2022

Built government services for millions of UK citizens. Ruby, Rails, PostgreSQL, GOV.UK PaaS.

- Built a service to **apply to court about child arrangements**, validated through extensive user research, improving access to justice for families.
- Led development of a **service to check when to disclose cautions or convictions**, translating complex legal requirements into custom algorithms with policymakers and legal experts.
- **Transformed court payment processing** with an online system that eliminated weeks-long delays, removed manual cheque handling, and enabled instant fund allocation to court budgets.

Senior Software Engineer, Creditspring

2018 – 2020

Joined a pre-launch fintech startup and architected scalable infrastructure, growing from 0 to 100K+ paying customers (500K+ registered users) while maintaining reliability.

- Architected **event-driven systems** using Rails, PostgreSQL, Redis, and Sidekiq, ensuring complete financial auditability and scalability for affordable borrowing products.
- Led the operations team to identify and implement automation, reducing manual processes to enable efficient scaling.
- Established monitoring and reliability practices maintaining **99.9%+ uptime** during rapid customer growth.

Software Engineer, Government Digital Service

2016 – 2018

Developed core GOV.UK features for 60M+ monthly visitors. Ruby, Rails, Elasticsearch.

- Led development of GOV.UK **Step by Step pages**, simplifying complex government processes into clear user journeys for millions of citizens.
- Designed and implemented a **taxonomy system** for GOV.UK content organisation, helping content creators improve information retrieval across departments.
- Enhanced GOV.UK **search with Elasticsearch**, optimising performance and accuracy for millions of daily searches and phasing out legacy systems.

EARLIER EXPERIENCE

Software Engineer roles at Unii, Sticky9, HouseTrip, and Think Orange. Further detail available on LinkedIn or on request.

PROJECTS & PUBLICATIONS

Modular Rails: Architecture for the Long Game — technical book on building modular monolith applications with Rails engines as a principled alternative to microservices; well received in the Ruby on Rails community.

Building Your Own Roguelike: A Practical Guide — technical book on game development and event-driven architecture in Ruby, with an accompanying open-source game.